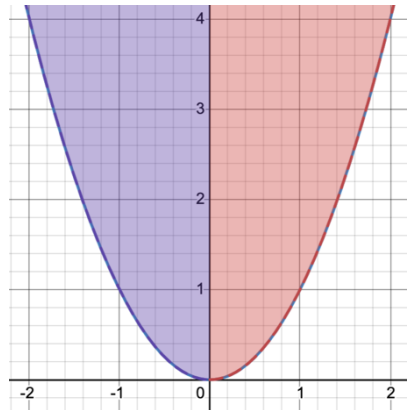


8.1 Identifying Key Features of Quadratic Functions

Lesson Introduction

 Benchmark	MA.912.AR.3.7 Given a table, equation, or written description of a quadratic function, graph that function and determine and interpret its key features.		 Learning Target	I can identify key features of a quadratic given a graph, a table, and an equation.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- How can I use different forms of quadratic equations to identify key features of a quadratic graph? - How are key features identifiable when given a table? - How are key features identified from a graph?			
 Lesson Narrative	In this lesson, students begin to explore the key features of a quadratic when given a graph, a table, and an equation. Students will explore x -intercepts, y -intercept, vertex, axis of symmetry, increasing and decreasing, and positive and negative key features of a quadratic equation.			
 Component Summary	8.1.1 Warm-Up (~5 min.)	Food Stylist career video (<i>SE p. 3</i>)	8.1.3 Collaboration (15-20 min.)	Exploration of key features of quadratics when given equations (<i>SE p. 5</i>)
	8.1.2 Guided Instruction (15-20 min.)	Introducing key features: vertex, x -intercepts, y -intercept, and line of symmetry (<i>SE p. 4</i>)	8.1.4 Guided Practice (10 min.)	Identify key features of quadratics using equations (<i>SE p. 8</i>)
	8.1.5 Wrap-Up (~5 min.)	Locate the key features on quadratic graph (<i>SE p. 9</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Vertex - Intercept - Line of Symmetry - Quadratic Function <u>Additional Terms:</u> N/A		- Food Stylist Career Video (Optional) Chart Paper and Markers (Optional) Colored Pencils or Highlighters	- Work through each activity on your own prior to providing instruction to highlight areas in which students may have misconceptions and need additional explanations, examples, or illustrations. 8.1.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions - Students may switch the values in an x-intercept and y-intercept. - Students may struggle to make connections between the forms of a quadratic equation and the key features it reveals. - Students may believe parabolas are always increasing or always decreasing.	Things to Consider Students first experienced quadratics in Grade 8 when solving $x^2 = p$. Students may benefit from Anchor Charts and reference notes to refer to as they continue through this unit and the remainder of the course.
	Literacy and Language Routines Notice and Wonder 8.1.3 Collaboration provides an opportunity to discuss what students notice and wonder about the values in a table, graph, and equation. Students will explore each representation and share what they notice and wonder.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students will be exposed to different representations of a quadratic: tables, graphs, and equations. They will work to express connections between the concept of key features and the different representations of quadratics in this lesson.
	 Differentiate Instruction	Consider using digital graphing technology, like Desmos or GeoGebra for exploring unique key features. For example, students can explore line of symmetry in this visual sample, making note of the colors red and purple, while you can toggle different colors on and off in real time. - What do you notice about the shaded area for each color? - What do you notice about the lines where they “meet” on the plane? - What do you wonder about the distance between parallel points on either side of the quadratic and that line of symmetry? For example, compare the points $(-2,4)$ and $(2,4)$. How are they the same? How are they different? What can this tell you about properties of a quadratic graph and its line of symmetry?
Lesson Notes:		



8.1.1 Warm-Up: Food Stylist

Component Overview: In this activity, students will explore the career of a Food Stylist. This activity is designed to be done independently.



8.1.1 Warm-Up: Food Stylist

Kim Kissling changed careers from working a corporate job to working as a food stylist after learning about the job during a career day at cooking school. As a food stylist, Kim takes photographs for cookbooks, magazines, advertising campaigns, product packaging, and television advertisements.



This Warm-Up activity has a video that can accompany the task. Consider whether students read the narrative paragraph before or after the video, depending on the learning preferences of your students.

1. What do you think would be interesting about working as a food stylist?

Answers will vary. I think it would be interesting to work as a food stylist because it is a creative avenue to work in photography and food, which are two of my favorite things. I think it would be interesting because it's a unique industry where lots of people would admire your work.

2. Kim said two important skills for her job are being super organized and not wasting time. How can you use these skills as a student?

Answers will vary. I think I can use these skills to be more effective in my studies. I can use these skills to get better grades in my classes.



- Do you have to be “super organized” and “not waste time” to be successful in math? Why or why not?
- Is this what it means to be “fluent” in something?

8.1.2 Guided Instruction: Key Features of Quadratics

Component Overview: Students will receive guided instruction on key features of quadratics, including: the vertex, x -intercept(s), y -intercept, intervals where the y -values are positive/negative, and the line of symmetry by exploring two graphed quadratic functions.



8.1.2 Guided Instruction: Key Features of Quadratics

The graph of a quadratic equation is called a parabola.

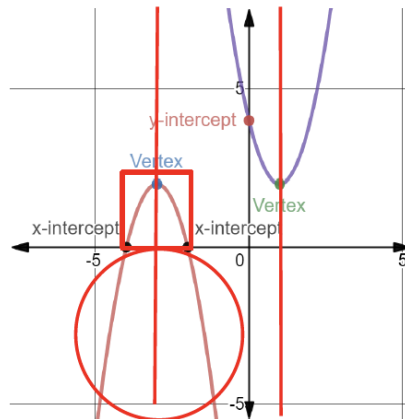
The **vertex** of a parabola is the point where the parabola changes direction.



The **intercept** is the value of a variable when all other variables in the equation equal 0. On a graph, the intercept is the value where a function crosses an axis.

1. Work with your partner to describe the red and purple parabolas. Note any similarities or differences.

Answers will vary. The red parabola has two x -intercepts, faces downwards, no visible y -intercept, the x -value of the vertex is negative, and the vertex is the highest point. The purple has no x -intercepts, the y -intercept is at $(0,4)$, it faces upwards, and the vertex is the lowest point. Both parabolas have a vertex, y -intercept, and a positive y -value of the vertex.



2. On the graph circle where the red parabola has negative y -values.

Answer on graph.



Leverage vocabulary strategies when introducing the key terms vertex, intercepts, and line of symmetry. For the intercept vocabulary, consider providing an example of a linear equation with an x -intercept and a y -intercept.



Allow students to have a conversation about question 1. Have students describe each parabola one at a time. Then, have students do comparisons between them based on what they see.

8.1.2 Guided Instruction: Key Features of Quadratics (continued)

3. Draw a square on the graph where the red parabola has positive y -values.

Answer on graph.

4. Discuss with your partner what key feature can be helpful in determining where a quadratic function has positive y -values and negative y -values.

Answers will vary. I think that any points below the x -intercepts would have negative y -values. I think that the x -intercepts are where the y -values equal 0, so anything that goes lower than that would have a negative y -value.

5. Discuss with your partner why the purple parabola only has positive y -values.

Answers will vary. The purple parabola has no x -intercepts, so it does not have any values that go below the x -intercept where the y -values would be negative. The purple parabola has no solution, so it stays above the x -axis the entire time, where y -values are only positive.



A **line of symmetry** is a line that divides a figure into two congruent parts, so that the reflection of either part across the line maps precisely onto the other.

6. Every parabola has a line of symmetry. Draw the line of symmetry for the red and purple parabolas.

Answers on original graph.

7. What do you notice about the two lines of symmetry?

Answers will vary. The red line has a line of symmetry on a negative x -value, while the purple line has a line of symmetry on a positive x -value. The line of symmetry cuts each parabola in half. The line of symmetry goes through the vertex.



When reviewing questions 2-5 as a class, consider doing a team activity review. Post the problem in a common area where all students can see (i.e., projector, display monitor, chart paper, whiteboard space, etc.). Have a student-partner group complete the steps on the common space for each question. Prior to moving on to the next question, challenge all students to determine additional information that may have been missed by the group. For questions 4 and 5, consider doing this by charting their answers on that same common space.



Depending on where the discussion leads, it would be interesting to ask specifically about the intercepts.

- x -intercepts are labeled on the red parabola. Do you think the purple parabola will have any x -intercepts?
- A y -intercept is labeled on the purple parabola. Do you think the red parabola has a y -intercept?



See the *Differentiate Instruction* section in the planning notes for strategies on instruction for line of symmetry.

8.1.3 Collaboration: Key Features of Quadratics Using Equations

Component Overview: Students will work collaboratively to explore the key features of quadratics when given equations in different forms. Begin this activity by reviewing the key features of increasing and decreasing using the narrative language (first paragraph).



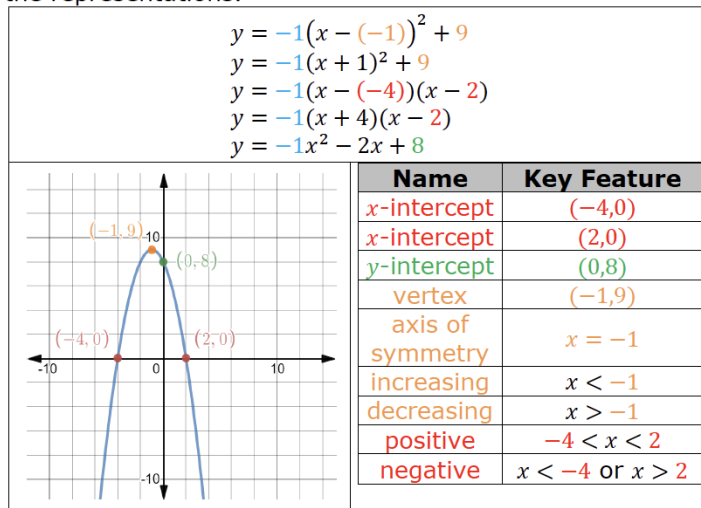
8.1.3 Collaboration: Key Features of Quadratics Using Equations

One of the key features studied with lines is whether the function is increasing or decreasing. Unless a line has a slope of 0 or an undefined slope, the line is always increasing or decreasing as x increases (from left to right).

- With your partner, use the red and purple parabolas to make a conjecture about where parabolas are increasing and decreasing.

Answers will vary. Parabolas like the red parabola will increase on the left side until the vertex, then decrease on the right side of the vertex. Parabolas like the purple parabola will be the opposite. They will decrease on the left side to the vertex, then increase on the right side of the vertex.

The table shows five forms of the same quadratic equation, $y = -1(x + 1)^2 + 9$, the graph of the quadratic, and a list of key features. The key features are color coded in each of the representations.



Consider creating an Anchor Chart or alternative method to chart the conjectures that students have about where the parabolas are increasing and decreasing. Allow all students to share their responses, placing checkmarks next to conjectures that students share. Revisit to debunk or verify conjectures throughout the lesson.



It may be beneficial to review the table of features and their corresponding values in both the table and the graph prior to releasing students to work on their first set of questions. Students may be confused by connecting the color in the equations to the color in the table and graph.



Consider Mark-Up Text strategies like colored highlighters, pencils, markers, or underlining and circling terms to highlight key features.



- What do you notice about the colors of the numbers in the equations and the colors of the key features from the table?

8.1.3 Collaboration: Key Features of Quadratics Using Equations (continued)

2. The equations $y = -1(x - (-1))^2 + 9$ and $y = -1(x + 1)^2 + 9$ are very similar. How can the vertex be determined when the equation $y = -1(x + 1)^2 + 9$ is given?

Answers will vary. The vertex can be determined by taking the opposite sign of the number in the parentheses as the x -value of the vertex and taking the number outside the equation at the end as the y -value of the vertex.

3. The equations $y = -1(x + 1)^2 + 9$ and $y = -1(x + 4)(x - 2)$ both represent the same quadratic. Rewrite both so that they are in the form $y = ax^2 + bx + c$. Show your work.

$y = -1(x + 1)^2 + 9$	$y = -1(x + 4)(x - 2)$
$y = -1(x + 1)(x + 1) + 9$	$y = -1(x^2 - 2x + 4x - 8)$
$y = -1(x^2 + x + x + 1) + 9$	$y = -1(x^2 + 2x - 8)$
$y = -1(x^2 + 2x + 1) + 9$	$y = -x^2 - 2x + 8$
$y = -x^2 - 2x - 1 + 9$	
$y = -x^2 - 2x + 8$	

4. The equations $y = -1(x - (-4))(x - 2)$ and $y = -1(x + 4)(x - 2)$ are equivalent. How can both x -intercepts be determined if the equation $y = -1(x + 4)(x - 2)$ is given?

Answers will vary. The x -intercepts can be determined by using the zero product property. I can set each factor equal to zero and then solve to determine the x -intercepts.

5. The equation $y = -1x^2 - 2x + 8$ can be used to determine the y -intercept. Show how each of the three forms can be used to find the y -intercept by substituting 0 for x .

$y = -1x^2 - 2x + 8$	$y = -1(x + 1)^2 + 9$	$y = -1(x + 4)(x - 2)$
$y = -1(0)^2 - 2(0) + 8$	$y = -1(0 + 1)^2 + 9$	$y = -1(0 + 4)(0 - 2)$
$y = 0 - 0 + 8$	$y = -1(1)^2 + 9$	$y = -1(4)(-2)$
$y = 8$	$y = -1(1) + 9$	$y = 8$
	$y = -1 + 9$	
	$y = 8$	



Continue facilitating opportunities for students to build their fluency in forms of quadratic equations.

- How is it possible for the equations to be written in different forms yet represent the same quadratic equation?
- Why do you think the different forms might be helpful when working with key features of quadratic equations?

8.1.3 Collaboration: Key Features of Quadratics Using Equations (continued)

6. A table of values for $y = -1x^2 - 2x + 8$ is shown.

- a. Check all the key features that can be found using the table of values.

- ☐ x -intercepts
- ☐ y -intercept
- ☐ vertex
- ☐ axis of symmetry
- ☐ increasing/decreasing
- ☐ positive/negative

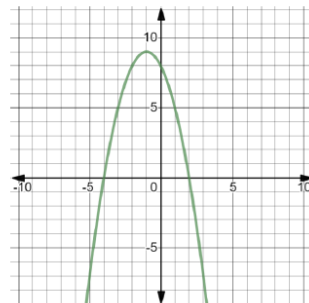
x	y
-3	5
-2	8
-1	9
0	8
1	5
2	0
3	-7

- b. Discuss with your partner the key features that were not found, but that could be found if the table of values included more values.

Answers will vary. I can't determine positive or negative because I am missing the 2nd x -intercept. I can deduce the direction of the parabola, but I need more points or the second x -intercept.

7. A graph of the function $y = -1(x - 2)(x + 4)$ is shown. Check all the key features that can be found using the graph.

- ☐ x -intercepts
- ☐ y -intercept
- ☐ vertex
- ☐ axis of symmetry
- ☐ increasing/decreasing
- ☐ positive/negative



If necessary, clarify that students need only to find one x -intercept to verify that x -intercepts can be found using a table of values.



For a challenge, consider having students take their answer from question 6a and their explanation from question 6b to create more values in the table that would allow them to be able to check off the remaining key features from the list.



Consider adding physical movement by assigning certain students to certain key features based on identifying factors like their birthday or first letter of their first name. For example, January-February birthdays are x -intercepts. Students then find a partner with a different birthday around the room and share their findings.

8.1.3 Collaboration: Key Features of Quadratics Using Equations (continued)

8. A table of values for $y = -\frac{2}{5}x^2 - 2x + 3$ is shown.

Check all the key features that can be found using the table of values.

- ☐ x-intercepts
- ☐ y-intercept
- ☐ vertex
- ☐ axis of symmetry
- ☐ increasing/decreasing
- ☐ positive/negative

x	y
-3	5.4
-2	5.4
-1	4.6
0	3
1	0.6
2	-2.6
3	-6.6

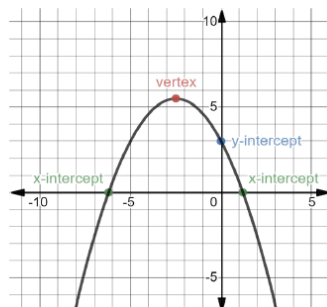


To add more challenge, have students add new values to the table in order to encompass all the key features that are not listed. At this stage, students may not be able to write the correct vertex or intercepts but look for their conceptual understanding of what the values would need to look like and do in order to meet the conditions for that specific key feature.

9. The equation $y = -\frac{2}{5}x^2 - 2x + 3$ is shown on the coordinate grid.

Which, if any, of the ordered pairs of the marked key features can be determined from the graph?

The y-intercept at (0,3).



Determining all of the key features of a quadratic may require using a graph, a different forms of an equation, and a table of values.



A **quadratic function** is a polynomial function with degree of 2.



Target fluency with the following questions:

- Are there certain key features that are easier to determine from a certain form over another?
- Which representation (graph, table, equation) do you find to be the easiest to work with when you are finding key features of quadratic equations?

8.1.4 Guided Practice: Key Features of Quadratics Using Equations

Component Overview: Students will engage in a structured, guided practice opportunity on identifying key features of quadratics using equations, tables, and graphs.

8.1.4 Guided Practice: Key Features of Quadratics Using Equations

Three different equations for a quadratic function are given.

- Use the equations to determine the key features. If an equation cannot be used to determine a key feature write "no equation."

Standard Form	Vertex Form	Factored Form
$y = 2x^2 + 8x - 24$	$y = 2(x + 2)^2 - 32$	$y = 2(x + 6)(x - 2)$

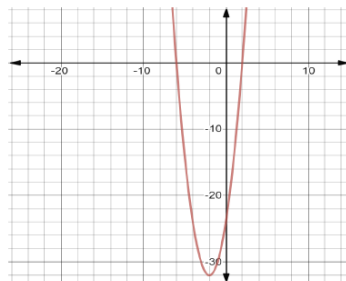
Name	Form Used	Key Feature
x-intercepts	<i>Factored Form</i>	$(-6,0), (2,0)$
y-intercept	<i>Standard Form</i>	$(0, -24)$
vertex	<i>Vertex Form</i>	$(-2, -32)$
axis of symmetry	<i>Vertex Form</i>	$x = -2$
increasing/ decreasing	<i>Vertex Form</i>	Increasing $\{x: x > -2\}$ Decreasing $\{x: x < -2\}$
positive/negative	<i>Factored Form</i>	Positive $\{x: x < -6 \text{ or } x > 2\}$ Negative $\{x: -6 < x < 2\}$

- Label the key features on the graph of $y = 2x^2 + 8x - 24$. Then, if possible, circle the key feature in the table of values.

Vertex: $(-2, -32)$

y-intercept: $(0, -24)$

x-intercepts: $(2,0)$ and $(-6,0)$



x	-3	-2	-1	0	1	2	3
y	-30	-32	-30	-24	-14	0	18



Consider using the *Collect and Display* instructional routine here. Call on students to provide a specified portion of the table, providing their rationale on why they selected that particular answer. Use a random name generator such as an online tool or the use of popsicle sticks to make the selection of students randomized for increased engagement.



Watch for student notation of the key feature column for increasing/decreasing and positive/negative intervals. Students may be able to articulate where these are occurring on the graph but may not recall how to properly notate it using inequalities. Draw specific attention to the language of that representation using inequalities.



For students who finish early, consider having them create a graph and trade with another student for them to locate what key features are present on their graph.

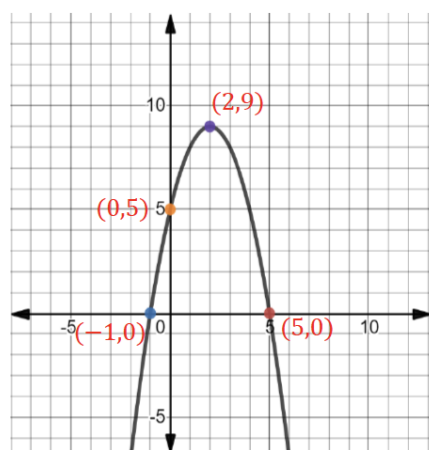
8.1.5 Wrap-Up: Cool Down

Component Overview: Students will complete a Cool Down that assesses a student's level of understanding of locating the key features on a quadratic graph. This activity is designed to be done independently to provide accurate student-level data.



8.1.5 Wrap-Up: Cool Down

- The graph of a quadratic equation is shown. Label the four colored points on the parabola with the ordered pair.



- Use words, ordered pairs, or symbols to describe the location of each key feature on the given parabola.

Key Feature	Location of Key Feature
x-intercepts	$(-1, 0), (5, 0)$
y-intercept	$(0, 5)$
vertex	$(2, 9)$
axis of symmetry	$x = 2$
increasing/decreasing	Increasing: $x < 2$ Decreasing: $x > 2$
positive/negative	Positive: $-1 < x < 5$ Negative: $x < -1$ or $x > 5$



Consider differentiating in Lesson 2 using students' responses in this table. For example, you can aggregate student data by feature.

- Identifying x-intercepts
- Identifying y-intercept
- Identifying the vertex
- Identifying the line of symmetry
- Identifying where a parabola is increasing and decreasing
- Identifying intervals where a parabola is positive and negative